

Twenty-five conjectures from a local–global random model, with machine verification and adversarial audit

The Conjecture Engine*

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Abstract

We state twenty-five conjectures in elementary and analytic number theory, each derived from the local–global random model of the primes: Cramér’s model corrected by Hardy–Littlewood singular series, with Bateman–Horn as the organizing framework and Borel–Cantelli accounting for sparse events. Each statement is calibrated to the strongest form the heuristic supports and no stronger. Every computable constant is computed from its definition, with local admissibility asserted programmatically; every statement was tested against exact counts, with success measured by the *shape* of agreement (predicted constant, square-root residuals, no drift) rather than by the size of the range searched. The statements were then subjected to a three-layer adversarial audit: an independent literature search for prior art, scripted refutation attempts a decade beyond the original bounds, and clean-context re-derivations of constants and counts. The audit refuted one conjecture as first stated—the exceptional set for $n = p + k^3$ contains the infinite family of cubes k^3 with $3k^2 - 3k + 1$ composite, an algebraic obstruction the probabilistic accounting missed—and it appears here restated, with the obstruction promoted to a theorem and the original error documented; a later round of review caught a priority failure (a statement we carried as new had been published), which is likewise documented rather than erased. We are deliberately conservative about novelty, and separate two kinds of content. A benchmark layer—eleven statements previously stated in essence and attributed, plus a handful of unrecorded but mechanical specializations of Bateman–Horn—is offered as calibrated test data rather than as new mathematics. The research group carries what we argue is conceptual novelty: a quantified Chebyshev-type race among twin-prime pairs modulo 5 and 8 whose deterministic term is driven entirely by the pairs $(q^2 - 2, q^2)$, together with a Goldbach lane race from the same mechanism (with a verified internal null lane) and a null-control race in which the mechanism is algebraically impossible and the predicted drift is zero; a finite-modulus deficit law for least primes in arithmetic progressions, with a reproducible ordering anomaly that persists on prime moduli, where smooth-modulus singular-series effects are absent; derived covariance kernels for the residual fields of the de Polignac and Hardy–Littlewood-F families (for the latter the computed cross-member kernel is null, localizing the observed sub-Poisson profile in the diagonal); a family-uniform quadratic de Polignac law; an alternating cyclotomic chain and the complete twin-base analysis of the quadratic cyclotomic family; the finiteness of factorial twins (the convergent-side Borel–Cantelli exemplar); a finite- x variance law for prime counts in microscopic windows with second-order constant $\gamma + \log 2\pi - 1$ (a microscopic extrapolation and

*This paper was produced by an automated conjecture engine (Claude, Anthropic), operated as an exercise in disciplined conjecture-making. All constants, counts, and audit results are reproducible from the accompanying repository (`conjecture-engine`): `run_all.py` regenerates every number in this paper. Primality of integers beyond 3.3×10^{24} was tested with Baillie–PSW; all smaller primality claims are deterministic (fixed-base Miller–Rabin).

test of Montgomery–Soundararajan at Gallagher’s boundary); an entanglement-aware local-global law for the mixed polynomial–exponential sequence $n^2 + 2^n$; a liminf first-occurrence law for prime gaps with the Granville-corrected constant $\sqrt{e^\gamma/2} = 0.943682\dots$, registered against Shanks’s classical constant 1; and the power-obstruction ladder for $n = p + j^k$ (total algebraic obstruction for composite k , a Bateman–Horn lane for prime k —the dichotomy itself corrected by a third external review, which refuted our first even/odd version).

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1 Introduction

A good conjecture in the theory of numbers is not merely a statement that has resisted counterexample. It is the *prediction of a model*: the assertion that the primes, beyond their local structure, behave like a random set of density $1/\log n$. The discipline this imposes is exacting. A conjecture worth stating should (i) survive every congruence and size obstruction, which is a computation, not a hope; (ii) come with a quantitative asymptotic whose constant is derived, not fitted; (iii) demand exactly the fluctuations the model earns—square-root noise with logarithmic corrections—and no fewer (the Mertens conjecture died of demanding fewer); (iv) sit inside the standard hierarchy of conjectures (Hardy–Littlewood k -tuples $\subset$ Schinzel’s Hypothesis H $\subset$ Bateman–Horn), so that its failure would propagate; and (v) be falsifiable instance-by-instance by computation. Numerical range is the least important column in the ledger: most deep phenomena drift at the rate $\log \log x$, and $\log \log 10^{18} \approx 3.7$, so verification “to 10^{18} ” is weak evidence by itself. What persuades is the *shape* of the agreement—a count that tracks a derived constant through several decades with residuals that look like noise.

This paper is the output of an automated engine built to those specifications, together with the audit that the specifications demand. The engine’s twenty-five statements are organized in four blocks: Bateman–Horn systems (§3), barely-divergent sparse sequences (§4), representation problems with Borel–Cantelli accounting (§5), and statistical laws of the prime sequence (§6). Section 7 describes the adversarial audit: what was attacked, what survived, and in one case what did not. We consider the audit trail part of the mathematical content. Its headline is instructive: the one outright refutation (Conjecture 13, first version) was produced not by bad luck but by a density-zero algebraic family—the cubes—invisible to a probability model that

integrates over density. This is precisely the classical lesson of admissibility, and the engine had to relearn it in public.

Two further audit outcomes deserve emphasis. First, an independent literature layer found that eleven of the twenty-five statements already exist in essence—some verbatim (Dubner’s twin-sum conjecture, the Stern list for $p + 2k^2$, Caldwell–Gallot’s $e^\gamma \log N$ laws), one with an exact constant catalogued as an OEIS entry (A188596, the sub-chain benchmark retained inside Conjecture 6)—and, in one instance caught only at the fourth round of review, a statement we had carried as new was found already published (Finding 2). We keep the attributed statements in the paper, restated in our uniform framework and re-verified at our bounds, with explicit attribution in §8; the residual claim to novelty of this paper rests on the research group below, each member of which has now been novelty-searched at the depth that the Finding 2 failure taught us to demand. Second, several of the engine’s *first drafts* were corrected by data before any external audit: the naive Poisson claim behind Conjecture 20 fails at every accessible scale (the corrected second-order law is the conjecture); the first-guess bias direction in the twin race of Conjecture 21 was reversed by a mod-3 computation; the naive Exp(1) mean in Conjecture 22 is off by a factor that took three controls to localize. We report these corrections because the difference between a fitted claim and a derived claim is exactly what an audit is for.

Taxonomy: benchmarks versus research conjectures

An external review of an earlier draft made a criticism we accept in full: an unrecorded specialization of Bateman–Horn is analogous to evaluating a classical special function at a new argument—worthwhile as data, but not a new conjecture in the conceptual sense. We therefore classify the twenty-five statements explicitly. *Benchmarks* (Conjectures 2, 3, 5, 7, 12–15, 17, 18, 23, together with the attributed cores of 1 ($d = 2$), 13, 16(i) and 24(i)): statements previously known in essence, or that follow mechanically from Hardy–Littlewood/Bateman–Horn once admissibility is checked; their role here is calibration—each carries a derived constant that the data must and do hit. *Research conjectures* (Conjectures 1 (the family law), 4, 6, 8, 9–11, 16(ii), 19–22, 24(ii), 25): statements involving a mechanism, law, family uniformity, or dichotomy we could not find in the literature, stated with their dependence assumptions explicit. This roster is the product of four successive external reviews, each of which raised the floor: slots whose occupants were judged bibliographically novel at most—or, in one case, found to be previously published (Finding 2)—were replaced or lifted to family level (the power-obstruction ladder 4, the complete twin-base family 5, the alternating cyclotomic chain 6, the null-mechanism race 8, the factorial twins 10, the Goldbach lane race 25, and the family lift of 1); every replacement was novelty-searched at abstract depth and verified before admission. Within the research group we regard the square-contamination mechanism family—21 and 25 with their negative control 8—together with Conjecture 22(ii) and the derived covariance kernels of 16(ii) and 24(ii) as the paper’s core, joined by 20 and 4; Conjectures 9, 11 and 19 carry open modeling questions flagged in place.

Verification standards

All counts below are exact (sieves; deterministic Miller–Rabin for $n < 3.3 \times 10^{24}$; Baillie–PSW beyond, flagged as such). Every comparison reports the Poisson-normalized residual $z = (\text{obs} - \text{pred})/\sqrt{\text{pred}}$. For sparse counts the cumulative comparison is contaminated at small x by the phantom mass of the main-term integral near its lower limit; in those cases we also

report per-decade increments, which are clean. Constants are computed by truncated Euler products with the truncation wobble (difference between cutoffs) reported as an error bar; for conditionally convergent quadratic and cubic products this wobble, not the last digit, is the honest precision.

2 Notation

p, q denote primes; γ is Euler's constant, $e^\gamma = 1.781072\dots$; φ is Euler's totient; $\phi = (1 + \sqrt{5})/2$; F_n the Fibonacci numbers; $p\#$ the primorial. For irreducible $f_1, \dots, f_k \in \mathbb{Z}[x]$ with positive leading coefficients, the Bateman–Horn singular series is

$$C(f_1, \dots, f_k) = \prod_p \frac{1 - \omega(p)/p}{(1 - 1/p)^k}, \quad \omega(p) = \#\{n \bmod p : p \mid f_1(n) \cdots f_k(n)\},$$

and the corresponding main term is

$$I(N) = \int_2^N \frac{dt}{\prod_i \log f_i(t)},$$

which absorbs the usual $1/\prod_i \deg f_i$. The Bateman–Horn conjecture [2] asserts $\#\{n \leq N : \text{all } f_i(n) \text{ prime}\} \sim C \cdot I(N)$ whenever $C \neq 0$; every conjecture in §3 is an instance with its constant evaluated. For nonlinear systems the product converges only conditionally; throughout, it is taken in the canonical ordering of increasing primes, in which convergence follows from the equidistribution of $\omega(p)$ about its Chebotarev mean. Our numerical truncation wobble is an error bar for this canonically ordered product, not a substitute for that convergence statement. We write $\text{Li}_k(x) = \int_2^x (\log t)^{-k} dt$. The twin singular series is $\mathfrak{S}(d) = 2C_2 \prod_{p|d, p>2} \frac{p-1}{p-2}$ for even d , with $2C_2 = 1.3203236\dots$; the Goldbach series $\mathfrak{S}(n)$ is the same product over odd $p \mid n$.

Admissibility ($\omega(p) < p$ for all p) is asserted programmatically during every constant computation. The check does real work: the natural-looking pair $\{n^2 + n + 1, n^2 + n + 3\}$ has $\omega(3) = 3$ and is rejected; the $k = 4$ branch of Conjecture 5 and the naive iterated chain behind Conjecture 6 were both rejected by the same machinery, and both rejections became part of the statements.

Novelty labels, from the independent literature audit (§8): (a) previously stated in essence; (b) classical family, quantified instance apparently unstated.

3 Bateman–Horn instances

The eight conjectures of this block share one hypothesis-generating move: choose an admissible system of polynomials, compute its singular series, and state the count. (Two members go further: 1 states a family-uniform law over all even shifts, and 8 is the mechanism family's negative control, a race prediction over the solutions of 1's system.) The heuristic is uniform, so we give it once. Model the primality of the values $f_i(n)$ as independent events of probability $1/\log f_i(n)$, corrected by the factor $(1 - \omega(p)/p)/(1 - 1/p)^k$ at each prime p to account for the joint local behaviour; multiplying the corrections and integrating gives $C \cdot I(N)$.

Conjecture 1 (Uniform quadratic de Polignac; family law apparently unstated, the $d = 2$ instance is (a), cf. OEIS A080149, [30]). *For even $d \geq 2$ let $\pi_d^q(N) = \#\{n \leq N : n^2 + 1, n^2 + 1 + d \text{ both prime}\}$ and $C(d) = C(x^2 + 1, x^2 + 1 + d)$. Then, for every fixed $B > 0$,*

$\pi_d^q(N) = C(d) I_d(N) (1 + o(1))$ uniformly over even $d \leq (\log N)^B$. In particular ($d = 2$, the classical instance that formerly stood alone in this slot): $\#\{n \leq N : n^2 + 1, n^2 + 3 \text{ both prime}\} \sim C(2) I(N)$, $C(2) = 2.954014 \dots$

This is the quadratic analogue of the uniform de Polignac law of Conjecture 16: not one count but the whole profile of constants must be matched at once, and the family, unlike its linear parent, has no known prior statement. Every even shift is admissible (for $p \geq 5$, $\omega(p) \leq 4 < p$; the classes mod 3 and 5 vary with d , which is what makes the $C(d)$ -profile nonconstant, spanning a factor ≈ 3 over $d \leq 300$). At $d = 2$ the local analysis reads: n even, $n \not\equiv 0 \pmod{3}$, $\omega(p) = 2 + \left(\frac{-1}{p}\right) + \left(\frac{-3}{p}\right)$ for $p \geq 5$, and the conjecture implies infinitely many twin pairs $(m + 1, m + 3)$ with m a perfect square. Profile verification at $N = 10^6$ over all 150 even shifts $d \leq 300$: correlation 0.99976 between observed and predicted counts, regression slope 1.0022, residual mean $z = +0.24$ with spread 0.84 and $\max_d |z| = 2.28$ —the shape-of-agreement standard applied across the family, not cherry-picked at one shift.

Conjecture 2 (Cubic shift; (b)). $\#\{n \leq N : n^3 + 2 \text{ prime}\} \sim C_2^* I(N)$, $C_2^* = C(x^3 + 2) = 1.298435 \pm 0.0003$.

For $p \equiv 2 \pmod{3}$ the cube map is a bijection mod p , so $\omega(p) = 1$ *exactly* and the local factor is $(1 - 1/p)^{-1}(1 - 1/p) = 1$: only $p \equiv 1 \pmod{3}$ (where $\omega \in \{0, 3\}$ according as -2 is a cubic residue) moves the product, which therefore converges only conditionally—hence the quoted wobble. No statement of Bunyakovsky type is proven for any cubic; the nearest theorem is Heath-Brown’s $x^3 + 2y^3$ [3], which is the partial-result neighbour that makes the single-variable statement a research programme rather than a monolith.

Conjecture 3 (Chernick chain; (a), Dubner [10]). $\#\{p \leq x : p, 2p - 1, 3p - 2 \text{ all prime}\} \sim C_3 I(x)$, $C_3 = C(x, 2x - 1, 3x - 2) = 2.858249 \dots$

The three primes form an arithmetic progression with common difference $p - 1$, first instance (3, 5, 7); for $p = 6m + 1$ the product $p(2p - 1)(3p - 2) = (6m + 1)(12m + 1)(18m + 1)$ is Chernick’s universal-form Carmichael number, so this conjecture quantifies the population from which the classical Carmichael constructions draw.

Conjecture 4 (The power-obstruction ladder: an elementary structural proposition with a Bateman–Horn corollary; apparently new as a family). For $k \geq 2$ and $m \geq 2$, write $D_k(m) = m^k - (m - 1)^k$, and call m^k representable if $m^k = p + j^k$ for some prime p and $j \geq 1$. Then: (i) [theorem] for composite k , no k -th power is representable: if r is a proper divisor of k , then for every $1 \leq j < m$ the difference $m^k - j^k$ has the proper factor $m^r - j^r > 1$ with cofactor > 1 ; (ii) [elementary] for prime k (including $k = 2$), m^k is representable if and only if $D_k(m)$ is prime; and D_k is irreducible for prime k (its roots $\zeta^j/(\zeta^j - 1)$, ζ a primitive k -th root of unity, form a single Galois orbit— D_k is a linear-fractional transform of the cyclotomic polynomial Φ_k); (iii) [conjecture] for each prime k the representable lane follows Bateman–Horn: $\#\{m \leq M : D_k(m) \text{ prime}\} \sim C_k \int_2^M dt / \log D_k(t)$.

Finding 1. Our first version of this ladder claimed an even/odd dichotomy, with a Bateman–Horn lane for every odd k . The third external review refuted it: for odd composite $k = rs$ the factorization in (i) applies (e.g. $D_9(m)$ is *never* prime), so the lane exists only for prime k . We verified the refutation directly ($k = 9, 15, 21, 25, 27, 33$: zero prime values of D_k to $m = 2000$, with the predicted divisor $D_r(m) \mid D_k(m)$ checked identically) and record that this is the second

algebraic-factorization failure the audit programme has caught in our own output—the exact failure mode the engine was built to detect, twice missed at generation time and twice caught downstream.

The family grew out of the audit refutation behind Conjecture 13 (the case $k = 3$). It replaces, at the suggestion of an external review, a bibliographically-novel-at-most quintuplet constellation that formerly occupied this slot (its data survive in the repository; its $4 \cdot 10^9$ verification, ratio 1.0004, remains part of the audit record as Hardy–Littlewood calibration).

Conjecture 5 (Twin cyclotomic bases: the complete quadratic family; no prior trace found). *For the three cyclotomic polynomials of degree 2 (indices k with $\varphi(k) = 2$, i.e. $k \in \{3, 4, 6\}$), consider $\Phi_k(n)$ and $\Phi_k(n + 1)$ simultaneously prime. Then: (i) [parity obstruction, elementary] for $k = 4$ one member of the pair $(n^2 + 1, n^2 + 2n + 2)$ is even for every n , so the only prime pair is $n = 1$, giving $(2, 5)$ —the same obstruction that forbids consecutive $n^2 + 1$ primes; (ii) [collapse, elementary] $\Phi_6(x) = \Phi_3(x - 1)$ identically, so the $k = 6$ pair at index n is the $k = 3$ pair at index $n - 1$: the family contains exactly one nontrivial statement; (iii) [conjecture, the live instance] there are infinitely many n for which the length-3 repunits in bases n and $n + 1$ are simultaneously prime: $\#\{n \leq N : \Phi_3(n) = n^2 + n + 1 \text{ and } \Phi_3(n + 1) = n^2 + 3n + 3 \text{ both prime}\} \sim C_5 I(N)$, with $C_5 = C(x^2 + x + 1, x^2 + 3x + 3)$.*

Both forms in (iii) are odd for every n (each is Φ_3 of an integer), and mod 3 each vanishes on exactly one class ($n \equiv 1$ and $n \equiv 0$ respectively), so the pair is admissible with $\omega(3) = 2$. Unlike the arbitrary shifted pair that formerly occupied this slot (replaced at the suggestion of an external review), the companion here is structurally forced: it is the same cyclotomic value at the consecutive base, making this the natural “twin” statement inside the cyclotomic-chain family of Conjecture 6. The family analysis (i)–(ii) is what a family *lift* of the statement uncovers, and both clauses were found by the engine’s own admissibility machinery: the $k = 4$ branch was rejected as inadmissible at $p = 2$ at singular-series time (both-odd count to 10^6 : zero, prime pairs: $n = 1$ only), and the $k = 6$ run returned counts *identical* to $k = 3$, which is the numerical shadow of the polynomial identity in (ii).

Conjecture 6 (The alternating cyclotomic chain; apparently new). *There are infinitely many primes p such that, writing $u = \Phi_3(p) = p^2 + p + 1$, the three integers*

$$p, \quad u = p^2 + p + 1, \quad \Phi_6(u) = u^2 - u + 1 = p^4 + 2p^3 + 2p^2 + p + 1$$

are all prime, with $\#\{p \leq x\} \sim C_6^{\text{ch}} I(x)$ for the Bateman–Horn system $\{x, x^2 + x + 1, x^4 + 2x^3 + 2x^2 + x + 1\}$, $C_6^{\text{ch}} = 3.6143 \pm 0.011$. The first chains are $(2, 7, 43)$ and $(3, 13, 157)$.

This is the repunit analogue of a Cunningham chain: a prime base p , its length-3 repunit $u = (p^3 - 1)/(p - 1)$, then $(u^3 + 1)/(u + 1)$. The chain must *alternate* between Φ_3 and Φ_6 : the naive iterate $\{p, \Phi_3(p), \Phi_3(\Phi_3(p))\}$ is *inadmissible*—if $p \equiv 2 \pmod{3}$ then $u \equiv 1 \pmod{3}$, so $3 \mid \Phi_3(u)$ always, while $p \equiv 1 \pmod{3}$ forces $3 \mid u$; the engine’s admissibility check rejected the naive chain at design time, and the alternating form is the unique degree- $\{1, 2, 4\}$ continuation that survives. The length-2 sub-chain $\{p, p^2 + p + 1\}$ is the classical cyclotomic Sophie Germain statement that formerly occupied this slot ($C = 1.521661 \pm 0.0005$, catalogue value 1.5217315..., OEIS A188596; it remains verified at $z = -1.06$ at 10^7 and is retained as calibration): infinitely many prime bases have prime repunits of length 3, the base-independent home of the phenomenon whose base-2 shadow is the Mersenne sequence. The length-3 chain stated here is, as far as we can find, unstated, and its quartic layer makes it the highest-degree system in the suite.

Conjecture 7 (Downward companion; (b); sequence is OEIS A062326). $\#\{p \leq x : p^2 - 2 \text{ prime}\} \sim C_7 I(x)$, $C_7 = C(x, x^2 - 2) = 3.383216 \dots$

Squares are $\equiv 0, 1 \pmod{3}$, so $3 \nmid p^2 - 2$ always—no loss at 3—and $\omega(q) = 1 + (\frac{2}{q})$ for the quadratic factor. The same pairs $(p, p^2 - 2)$ reappear as the deterministic term of the twin race in Conjecture 21: this conjecture’s counting function sets that term’s magnitude. (The dependence is one-directional and partial—a race bias could conceivably be established in averaged form without the full asymptotic here—but a refutation of this conjecture would remove Conjecture 21’s predicted mechanism.)

Conjecture 8 (The null-mechanism race; apparently new). *For the quadratic twin pairs of Conjecture 1 at $d = 2$, square contamination is algebraically impossible: $n^2 + 1 = q^2$ has no solution with $n \geq 1$, and $n^2 + 3 = q^2$ only at $(n, q) = (1, 2)$. The surviving classes of $n \bmod 5$ are $\{0, 1, 4\}$, and the classes 1 and 4 have identical singular series (elementary, via CRT). Let $\pi_a^q(x)$ count solutions with $n \equiv a \pmod{5}$ and $D(x) = \pi_1^q(x) - \pi_4^q(x)$. Then, in contrast to the twin race of Conjecture 21, this race is driftless: the normalized difference $D(x)/\sqrt{\pi_1^q(x) + \pi_4^q(x)}$ has logarithmic mean tending to 0, and no clause analogous to Conjecture 21(i) holds—there is no deterministic term at the contamination scale, and leadership wanders in arcsine fashion with no persistent leader.*

This is the negative control that completes the mechanism family of Conjectures 21 and 25: those races carry a predicted drift *because* prime squares can infiltrate specific residue classes of specific patterns; here the quadratic form blocks squares identically, so the same model predicts *nothing*—a falsifiable contrast. If a persistent drift were found in this race, the mechanism story behind the family would be wrong no matter how well the positive cases fit. Verification at 10^7 : 32,898 pairs, class counts 10,917 against 10,981 ($D = -64$, i.e. -0.43 noise units), log-mean normalized drift -0.46 (consistent with 0 at one noise unit), weighted lead fraction 0.21—deep in arcsine territory, exactly the profile of a fair race. (The quadratic triple $n^2 + \{1, 3, 7\}$ that formerly occupied this slot, $C = 10.64599 \pm 0.006$, is retired to the audit record with its calibration intact—observed 3,963 against predicted 3,998.6 at 10^7 , $z = -0.56$, independently recomputed to 3×10^7 ; the third external review found the longer pattern $n^2 + \{1, 3, 7, 9, 13\}$ recorded in the OEIS since 2000, so the tuple family has prior presence and only its singular-series evaluation was plausibly new.)

Verification

	bound	observed	predicted	ratio	z
C1 _{$d=2$}	10^7	32,898	32,862.7	1.0011	+0.19
C2	10^7	287,956	287,784.8	1.0006	+0.32
C3	$3 \cdot 10^8$	125,379	125,429.4	0.9996	−0.14
C4 _{$k=2$}	10^7	1,270,606	1,270,902.8	0.9998	−0.26
C4 _{$k=3$}	10^6	126,826	126,641.6	1.0015	+0.52
C4 _{$k=5$}	$8 \cdot 10^5$	56,925	57,021.3	0.9983	−0.40
C5	10^7	33,274	32,976.7	1.0090	+1.64
C6	10^7	1,362	1,357.2	1.0035	+0.13
C7	10^7	74,914	75,273.8	0.9952	−1.31

For C1, the family-profile verification (150 shifts) is reported at the statement; the table row is the $d = 2$ instance. For C4: the even-rung theorem was checked numerically to 10^8 (no

fourth power is $p + j^4$); the three verifiable Bateman–Horn lanes $k = 2, 3, 5$ (constants 2 exactly, 3.36181, and 3.67770 ± 0.007 , the quartic product computed by brute-force root counts to 10^5) all sit within $|z| < 0.6$. For C5, $C_5 = 2.964239 \pm 0.002$. For C6, the chain census by decade is 4, 11, 41, 224, 1362 at $N = 10^3, \dots, 10^7$ (z from +0.40 to +0.13, no drift), the quartic layer’s singular series computed by brute-force root counts to 10^5 ; the retained sub-chain benchmark $\{p, p^2 + p + 1\}$ stands at 33,661 against 33,855.5 at 10^7 ($z = -1.06$). For C8, the race data are reported at the statement. Independent clean-context recomputation gave $C_7 = 3.383227$ and, for the retired occupants of slots 6 and 8 (the Sophie Germain sub-chain and the quadratic triple), 1.521730 and 10.647706, with primes to 10^9 (each inside our stated wobble) and independent counts to 3×10^7 with final ratios 0.9962, 0.9965, 1.0059. (The quintuplet that formerly occupied slot 4 retains its audit-record calibration: observed 15,236 against predicted 15,230.3 at $4 \cdot 10^9$, ratio 1.0004, $z = +0.05$.)

4 Sparse sequences: barely-divergent Borel–Cantelli

For sequences whose n th term has size e^{cn} the expected number of primes is $\sum_n \kappa_n / (cn)$: convergence or divergence of this sum is the entire question, and when it diverges it diverges logarithmically, so the right conjecture has counting function $\alpha \log x$ with a derived α —never more.

Conjecture 9 (Fibonacci primes; cf. [7]). $\#\{p \leq x : F_p \text{ prime}\} \sim c_F \log x$ for a constant $c_F > 0$ determined by the local (rank-of-apparition) structure of the Fibonacci sequence in the sense of the Grantham–Granville recurrence-sequence model. The first-order screening value is $c_F^{(0)} = e^\gamma / \log \phi = 3.70122\dots$; the data below indicate $c_F < c_F^{(0)}$, and we deliberately do not conjecture the exact constant.

Every prime factor of F_p ($p \neq 5$ prime) has rank of apparition exactly p , hence is $\equiv \pm 1 \pmod{p}$: the Lenstra–Pomerance–Wagstaff screening [8] transfers with $\log 2 \mapsto \log \phi$. The data sit persistently *below* the constant: 25 Fibonacci prime indices $\leq 10^4$ (independently reconfirmed by a second implementation) against a predicted 34.1, ratio ≈ 0.73 , $z = -1.6$, stable over three decades. Grantham and Granville [7] caution that naive constants for such sequences require $O(1)$ local corrections; the mod-5 divisor structure of F_p is the natural suspect, and we flag this constant as the component of the suite most likely to need a Granville-type patch.

Conjecture 10 (Factorial twins; apparently new). *Only finitely many n have $n! - 1$ and $n! + 1$ simultaneously prime, and the complete list is $n = 3$: i.e. $6 = 3! = 3\#$ is the unique factorial lying between twin primes.*

Each event separately has Caldwell–Gallot probability $\sim e^\gamma / n$ [4] and divergent sum (the $n! + 1$ side is exactly Conjecture 12); the *joint* event has probability $\sim (e^\gamma / n)^2$, whose sum converges. This is the suite’s exemplar of the convergent side of the Borel–Cantelli dichotomy—the accounting that predicts finitude rather than infinitude. Verified for $n \leq 700$; the model-expected number of further examples beyond 700 is $e^{2\gamma} / 700 \approx 4.5 \times 10^{-3}$ (computed tail). The defensible core is finiteness; the exact one-element list is the Goldilocks-maximal form—strictly stronger than the tail estimate can guarantee, and chosen deliberately as the most falsifiable statement.

Finding 2. The statement that previously occupied this slot—primorial twins: $p\# \pm 1$ both prime only for $p \in \{3, 5, 11\}$ —was carried for three revisions under an “apparently new” label, and

the fourth external review found it stated in Lillie [5], whose abstract gives both the $O(n^{-2})$ joint probability and the prediction that there are in total three instances. The failure is instructive and we record it against ourselves: our novelty search had *surfaced that exact preprint*, but the search snippet described only its single-sided results, and we did not read to the abstract's second half. Priority checks on a conjecture must read the abstract and body of near-neighbour papers, not their search summaries; the audit protocol now requires it. The primordial statement remains, attributed to Lillie, as calibration (our independent verification to $p \leq 4000$ stands), and the slot now carries the factorial analogue, for which our deeper search—including the factorial-prime literature and the Lillie preprint itself—found no prior statement.

Conjecture 11 ($n^2 + 2^n$; sequence is OEIS A064539). *Every prime of the form $n^2 + 2^n$ with $n > 1$ has $n \equiv 3 \pmod{6}$ (elementary), and: (i) infinitely many such primes exist, with counting function $\asymp \log N$; (ii) [candidate law, entanglement-aware form] for a finite set S of primes let D_S be the exact joint density, within the lane $n \equiv 3 \pmod{6}$, of n with $p \nmid n^2 + 2^n$ for all $p \in S$, computed over the full joint period $\text{lcm}_{p \in S} \text{lcm}(6, p \cdot \text{ord}_p 2)$, and set $\kappa_S = D_S / \prod_{p \in S} (1 - 1/p)$. The conjecture is that the net $\{\kappa_S\}$ converges as S increases to all primes, with limit κ , and that the count is $\sim \sum_{n \leq N, n \equiv 3 \pmod{6}} \kappa / \log(n^2 + 2^n)$. The working value, from the CRT-exact core $p \leq 19$ times per-prime factors to $p \leq 300$, is $\kappa = 4.2734 \dots$*

This is not a Bateman–Horn problem: the local conditions at different primes are tied through the orders $\text{ord}_p 2$, and the per-prime product $\prod_p (1 - \delta_p)/(1 - 1/p)$ of an earlier draft tacitly assumed their independence. Following external review, the law is now stated through the CRT-exact quantities κ_S , which make no independence assumption. The computation then returned a small surprise in the other direction: through $p \leq 19$ (joint period 116,396,280) the exact joint density *factorizes to machine precision*— κ_S agrees with the naive per-prime product to 10^{-15} at every level $S = \{p \leq P\}$, $P \in \{5, 7, 11, 13, 17, 19\}$. The entanglement the reviewer warned about is thus computably absent at all accessible levels; whether exact factorization persists for all p (which would itself be a striking equidistribution statement about the orbits of 2) is part of what clause (ii) asks. We separate the robust claim (i) from the candidate law (ii), and note that the eight hits below 6000 are far too few to validate a four-digit constant—the verification below checks consistency, not the constant.

The congruence claim is elementary: n odd is forced by parity, and for odd n with $3 \nmid n$ we have $n^2 + 2^n \equiv 1 + 2 \equiv 0 \pmod{3}$. What appears to be a harsh obstruction is, inside the surviving lane, a local bonus: $\delta_2 = \delta_3 = 0$ contribute the factors $2 \cdot \frac{3}{2} = 3$ to κ . Mixed polynomial–exponential sequences are exactly where hand-waving fails and the computed local product earns its keep.

Conjecture 12 (Factorial primes; (a), [4]). $\#\{n \leq N : n! + 1 \text{ prime}\} \sim e^\gamma \log N$.

Verification

C9: hits $\{3, 5, 7, \dots, 9311, 9677\}$, 25 of predicted 34.1 ($z = -1.56$; see the remark above). C10: exhaustive to $n \leq 700$: the only factorial twin is $n = 3$; individually, $n! + 1$ is prime for $n \in \{2, 3, 11, 27, 37, 41, 73, 77, 116, 154, 320, 340, 399, 427\}$ and $n! - 1$ for $n \in \{3, 4, 6, 7, 12, 14, 30, 32, 33, 38, 94, 166, 324, 371\}$, two divergent single-sided laws straddling their shared constant while the joint count stops at one, the convergent accounting in action. (The primordial companion, now attributed to Lillie [5] per Finding 2: exhaustive to $p \leq 4000$, twins exactly $\{3, 5, 11\}$, with $p\# + 1$ prime for eleven p against the Caldwell–Gallot prediction 14.8, $z = -0.98$.) C11: hits $\{3, 9, 15, 21, 33, 2007, 2127, 3759\}$ up

to 6000: observed 8, model 8.55, $z = -0.19$ (at the earlier bound 4200: 8 vs 8.19, $z = -0.07$); the lane constraint was verified exhaustively (no prime off $n \equiv 3 \pmod{6}$ to $n = 3000$, independently reconfirmed), and the CRT-exact factorization of clause (ii) was verified through $p \leq 19$ as described at the statement. C12: 15 primes with $n \leq 700$ against 11.7 ($z = +0.98$), completing the $e^\gamma \log$ family benchmark that Conjecture 10 joins on its convergent side.

5 Representation problems

Theorem 1 (Cube obstruction; found by the adversarial audit). *For $k \geq 2$, the cube k^3 is representable as $p + j^3$ with p prime, $j \geq 1$, if and only if $3k^2 - 3k + 1$ is prime. Consequently the set of integers not representable as $p + k^3$ is infinite: the non-representable cubes alone have counting function $\sim x^{1/3}$. (That the full exceptional set is $\sim x^{1/3}$ follows only when combined with Conjecture 13(i), and is conjectural.)*

Proof. $k^3 - j^3 = (k - j)(k^2 + kj + j^2)$; for $1 \leq j \leq k - 2$ both factors exceed 1, so the difference is composite. The only candidate is $j = k - 1$, giving $3k^2 - 3k + 1$. Since $3k^2 - 3k + 1$ is composite for a density-one set of k (all but $O(K/\log K)$ of $k \leq K$, by an upper-bound sieve of Brun or Selberg type applied to the quadratic's prime values), all but a vanishing proportion of cubes are unrepresentable. $\square$

Finding 3. As first generated (“the exception set of $n = p + k^3$, $k \geq 1$, is finite”), the next conjecture was **false**, and the engine’s own adversarial battery refuted it: all 412 exceptions found in $(10^8, 10^9]$ are perfect cubes, explained by Theorem 1. The error is a textbook violation of the admissibility clause of the design criteria—an algebraic obstruction along a density-zero subsequence, invisible to a Borel–Cantelli sum taken over all n . We record the failure rather than hide it; the restated version separates theorem from conjecture.

Conjecture 13 (Prime plus a cube, restated; (a)-core: the non-cube finiteness is Hardy–Littlewood’s $E_3(X) = O(1)$ [1]). (i) *The set of non-cube integers $n \geq 2$ not representable as $p + k^3$ ($k \geq 1$) is finite.* (ii) *The representable cubes obey Bateman–Horn: $\#\{k \leq K : 3k^2 - 3k + 1 \text{ prime}\} \sim C(3x^2 - 3x + 1) I(K)$.*

(Empirically the non-cube exception census by decade is 4, 27, 168, 763, 2011, 2808, 1181, 88 up to 10^8 , peaking at the sixth decade and collapsing thereafter, with largest element found 78,526,384; these are verification data, not part of the formal statement.)

Conjecture 14 (Stern list; (a): OEIS A060003). *Exactly ten odd integers are not of the form $p + 2k^2$ with $k \geq 1$:*

$$1, 3, 17, 137, 227, 977, 1187, 1493, 5777, 5993.$$

The expected number of representations of odd n is $\asymp \sqrt{n}/\log n$, so the expected number of exceptions beyond any X decays like $\sum_{n>X} \exp(-c\sqrt{n}/\log n)$: the list should be complete, and the seven prime members are precisely the Stern primes. Verified here to 10^9 , one decade past our original bound, with no eleventh exception.

Conjecture 15 (Goldbach in the (3, 3) lane; (a), K. Martin [11]). *Every $n \equiv 2 \pmod{4}$ with $n \geq 6$ is $p + q$ with $p \equiv q \equiv 3 \pmod{4}$, and the ordered representation count obeys*

$$R_3(n) \sim \frac{1}{2} \mathfrak{S}(n) \int_3^{n-3} \frac{dt}{\log t \log(n-t)}.$$

Normalization: $R_3(n)$ counts *ordered* pairs (p, q) , and the full-interval integral is the ordered main term for all Goldbach representations, so the factor $\frac{1}{2}$ is exactly the equidistribution split between the (1, 1) and (3, 3) lanes (parity forces $p \equiv q \pmod{4}$ when $n \equiv 2 \pmod{4}$). The (3, 3) lane keeps its smallest representations $(3 + q)$ and is the lane with *no* exceptions at all, which is why the uniform threshold $n \geq 6$ can be conjectured. Verified exhaustively to 10^9 ; the count formula matches to 0.2–1.3% at $n \approx 10^6, 10^7, 10^8$.

Conjecture 16 (Uniform quantitative de Polignac; (a)-core). (i) $\pi_d(x) = \#\{p \leq x - d : p, p + d \text{ prime}\} = \mathfrak{S}(d) \text{Li}_2(x) (1 + o(1))$ uniformly for even $d \leq (\log x)^A$. (ii) [residual law, with derived covariance kernel] With $z_d(x) = (\pi_d(x) - \mathfrak{S}(d) \text{Li}_2(x)) / \sqrt{\mathfrak{S}(d) \text{Li}_2(x)}$: the same primes participate in many π_d , and the covariance of two pair counts is carried by triple configurations sharing one prime. For even $d \neq d'$ define the overlap constant

$$K(d, d') = C_3(0, d, d') + C_3(0, d', d + d') + C_3(0, d, d + d') + C_3(0, |d - d'|, \max(d, d')),$$

where C_3 denotes the Hardy–Littlewood triple constant of the offset set (degenerate or inadmissible triples contributing 0). Then $\text{Cov}(\pi_d, \pi_{d'}) = K(d, d') \text{Li}_3(x) (1 + o(1))$, so the correlation kernel is $\rho(d, d') \sim K(d, d') \text{Li}_3(x) / (\mathfrak{S}(d) \mathfrak{S}(d'))^{1/2} \text{Li}_2(x)$, and the conjecture is that the residual field $\{z_d(x)\}_{2|d \leq 2D(x)}$, $D(x) = (\log x)^A$, converges to a centered Gaussian field with this covariance, with no trend in d . The diagonal (4-tuple) correction to $\text{Var}(\pi_d)$ below its Poisson value—the pair analogue of the Montgomery–Soundararajan deficit of Conjecture 20—is the remaining underived component, and we state it as open rather than fold it into a fitted σ .

Numerical evaluation of the kernel (all even $d, d' \leq 60$ at $x = 10^8$, triple constants by Euler products to 5×10^4): mean off-diagonal $\rho = 0.43$, hence predicted common-mode variance $\text{Var}(\bar{z}) \approx 0.41$ and predicted profile spread $\sigma^2 \leq 1 - 0.41 = 0.59$ before the diagonal correction; the observed profile variance ≈ 0.27 at $x = 10^8$ then implies a diagonal (sub-Poisson) factor ≈ 0.5 , of the same sign and order as the Montgomery–Soundararajan mechanism supplies for single primes. An earlier draft asserted only “ $\sigma^2 \leq 1$, determined by the covariance structure”; the external review rightly called that a task rather than a law, and the kernel above is the law the task produces.

The force of the statement is its uniformity: the singular series takes a thousand different values over $d \leq 2000$, spanning a factor of about three between $d = 2$ and the primorial-rich d , and the counts must reproduce the entire profile simultaneously with square-root errors. A conspiracy would need to bend a thousand-dimensional vector, not one number. At $x = 10^8$: correlation 0.999999, slope 0.99966, $\max_d |z_d| = 1.80$; stressed to $d \leq 6000$, $\max |z_d| = 2.01$ over three thousand values.

Conjecture 17 (Dubner’s conjecture; (a) [9]). *Every even $n \geq 4210$ is a sum of two members of twin-prime pairs; the 35 exceptions (OEIS A007534, largest 4208) are complete.*

Twin members have density $\asymp 1/\log^2 x$, so expected representations grow like $n/\log^4 n$ and exceptions must be finite; the observed exceptions cluster in triples of consecutive even numbers, a structural fingerprint of the sparse low-lying twin population. Verified to 10^8 , adversarially re-swept to 10^9 , and independently re-verified to 10^9 by a second implementation with a different algorithm: the same 35 exceptions each time.

6 Statistical laws

Conjecture 18 (Twin-gap records; (a), Kourbatov [13]). (i) *The maximal gap $G_t(x)$ between consecutive twin-prime starts up to x satisfies $G_t(x) \asymp \log^3 x$.* (ii) *[candidate constant] $\limsup_x G_t(x)/\log^3 x = 1/(2C_2) = 0.757390\dots$ Part (i) is the robust claim; the exact constant in (ii) inherits the known fragility of Cramér-style extreme-value constants (Granville’s correction arose precisely because extreme events feel local sieve scales invisible to an inhomogeneous Poisson model), and is offered as a working value, not an assertion at the same confidence as (i).*

Twin starts thin the integers to density $2C_2/\log^2 t$; a Cramér process at that density has record waiting times $\sim (\text{mean spacing}) \times \log(\text{number of events}) = \log^3 x/2C_2$. The Cramér–Granville episode teaches that such constants can be shifted by local corrections (Granville’s $2e^{-\gamma}$ [15]); the claim to take seriously is the cube of the logarithm, and the constant is flagged as first-order. Records to $4 \cdot 10^9$ (largest observed twin-to-twin gap 5292, following the twin at 2,466,641,069) wander in $[0.41, 0.56] \cdot \log^3 x$, approaching the asymptote from below on a log log clock, as such statistics must.

Conjecture 19 (First occurrences of prime gaps, liminf form; (b)). *Let $\mathcal{G}$ be the set of even g that occur as gaps between consecutive primes (all even g , under Polignac’s conjecture), and for $g \in \mathcal{G}$ let $p(g)$ be the prime starting the first such gap. Then: (i) $\log p(g)/\sqrt{g}$ is bounded between positive constants on $\mathcal{G}$; (ii) [liminf law]*

$$\liminf_{\substack{g \rightarrow \infty \\ g \in \mathcal{G}}} \frac{\log p(g)}{\sqrt{g}} = \sqrt{e^\gamma/2} = 0.943682\dots,$$

the liminf being attained along gap sizes realized at the Cramér–Granville maximal-gap envelope. (The restriction to $\mathcal{G}$ repairs a definitional defect caught by external review: $p(g)$ is not known to be defined for every even g , and quantifying over all g would smuggle Polignac’s conjecture into the statement.)

An earlier draft asserted the full limit, inverting Granville’s $\limsup G(x)/\log^2 x = 2e^{-\gamma}$; an external review correctly observed that this inversion is invalid. A limsup envelope for maximal gaps controls only where gaps *can first appear at the earliest*, i.e. a lower envelope for $p(g)$ —hence a liminf claim, not a limit. Whether $\log p(g)/\sqrt{g}$ converges at all, and if so whether the limit is Shanks’s classical 1 [14] (as Wolf’s refinement $p(g) \sim \sqrt{g}e^{\sqrt{g}}$ and the Kourbatov–Wolf residue-class laws [23] also assert) or the envelope value, involves at least three distinct phenomena—whether a given g occurs near the extreme scale at all, the distribution of exact gap lengths near records, and the difference between record gaps and non-record first occurrences—and is left open here. The two candidate constants differ by 6%, beneath the resolution of any feasible computation: our regression slope over $g \geq 100$ falls from 1.307 at 10^9 to 1.297 at $4 \cdot 10^9$, far above both and decreasing, as it must be for either.

Conjecture 20 (Microscopic variance law; (b); the constant is Montgomery–Soundararajan’s [17]). *Fix $\lambda > 0$ and $c \in (0, 1]$, and let X count primes in $[t, t + \lambda \log x)$ for t uniform on $[x, (1+c)x]$. Then, under the Hardy–Littlewood k -tuple conjectures, $X \Rightarrow \text{Poisson}(\lambda)$ (Gallagher’s argument [16], which is conditional on exactly those conjectures), and at finite x*

$$\frac{\text{Var } X}{\mathbb{E} X} = 1 - \frac{\log(\lambda \log x) + \gamma + \log 2\pi - 1}{\log x} + o\left(\frac{1}{\log x}\right),$$

with $\gamma + \log 2\pi - 1 = 1.41509\dots$, the leading correction being independent of the sampling constant c .

The deficit arises by inserting Montgomery’s singular-series average $\sum_{d \leq h} \mathfrak{S}(d)(h - d) = \frac{h^2}{2} - \frac{h}{2}(\log h + \gamma + \log 2\pi - 1) + o(h)$ into the second factorial moment at the microscopic window $h = \lambda \log x$. We state plainly, following external review, what this is and is not: the constant $\gamma + \log 2\pi - 1$ is Montgomery–Soundararajan’s, was identified in their framework long before this paper, and the statement is a *microscopic extrapolation and numerical test* of their second-moment formula at Gallagher’s boundary—not a newly discovered second-order law. The hypothesis must also be quantitative: what is required is not the qualitative k -tuple conjectures but uniform Hardy–Littlewood estimates for all shifts up to $O(\log x)$, control of the average singular series at the $o(h)$ level, and bounds for prime powers, endpoint weights, and the variation of $\log t$ across the sampling range. Three further cautions are built into the statement: Gallagher’s Poisson law is conditional, not a theorem about primes; the sampling measure for t is specified, since different weightings shift lower-order terms; and the error is claimed only as $o(1/\log x)$. The naive claim $\text{Var}/\mathbb{E} = 1$ fails at every accessible scale by 18–28%; the corrected law matches with no fitted parameter, and the observed residual $\approx +0.003$ is consistent with (but not evidence for) a genuine $1/\log^2 x$ term. (Numerical precedent: Sanchis-Lozano [24] fitted the Montgomery–Soundararajan variance shape at $x \sim 10^8$ in the mesoscopic regime $h \gg \log N$; the microscopic statement and test appear to be new, and we claim no more than that.)

	$x = 10^9$		$x = 4 \cdot 10^9$	
λ	observed	predicted	observed	predicted
1/2	0.8241	0.8189	—	—
1	0.7910	0.7854	0.7998	0.7960
2	0.7527	0.7520	0.7670	0.7646
4	0.7201	0.7185	—	—

The residual $\approx +0.003$ is the anticipated $O(1/\log^2 x)$ term.

Following the fourth external review, the twin-race statement is now split into its provable algebraic core and its conjectural clauses, with the fluctuation functional defined rather than gestured at. For a locally integrable F write $\mathcal{M}_x(F) = \frac{1}{\log x} \int_2^x F(t) \frac{dt}{t}$ for the logarithmic mean.

Lemma 1 (Orientation elimination and class assignment). *In $\psi_2(x) = \sum_{n \leq x} \Lambda(n)\Lambda(n+2)$, the total contribution of terms in which n or $n+2$ is a proper prime power is, apart from $O(x^{1/3} \log^2 x)$ from cubes and higher powers, carried entirely by the patterns $(q^2 - 2, q^2)$ with q prime and $q^2 - 2$ prime: the mirror pattern $(q^2, q^2 + 2)$ is annihilated by $3 \mid q^2 + 2$ for every prime $q > 3$. Moreover, for every prime $q \neq 5$, $q^2 - 2 \equiv 2$ or $4 \pmod{5}$ (never 1), and for every odd q , $q^2 - 2 \equiv 7 \pmod{8}$. (The single exceptional term $q = 5$, $(23, 25)$, lands outside the twin-start classes altogether and is absorbed in the bounded error.)*

The proof is elementary (squares mod 3, 5, 8; the prime-power count). The lemma orients the race: prime-square contamination can enter only specific residue classes, and the model converts that into a drift prediction for the *unweighted* twin counts.

Conjecture 21 (Twin race modulo 5 and 8; (b), apparently new). *Twin starts $p > 5$ lie in classes $p \equiv 1, 2, 4 \pmod{5}$. Let $D_1(x) = \pi_t(x; 5, 1) - \frac{1}{2}(\pi_t(x; 5, 2) + \pi_t(x; 5, 4))$ and*

$$T(x) = \frac{1}{2 \log^2 x} \sum_{\substack{q \leq \sqrt{x} \\ q^2 - 2 \text{ prime}}} \log q \log(q^2 - 2),$$

whose scale is governed by the constant C_7 of Conjecture 7. Then: (i) [drift law] class 1 carries the systematic surplus T :

$$\mathcal{M}_x\left(\frac{D_1 - T}{\sqrt{\pi_t}}\right) \rightarrow 0 \quad (x \rightarrow \infty),$$

i.e. the deterministic part of D_1 is exactly T , and the remainder is mean-zero noise at the random-walk scale under logarithmic averaging; (ii) [null covariance] the symmetric classes carry no deterministic term: $\mathcal{M}_x((\pi_t(\cdot; 5, 2) - \pi_t(\cdot; 5, 4))/\sqrt{\pi_t}) \rightarrow 0$, and likewise for each of the three pairwise differences among classes $\{1, 3, 5\}$ mod 8 in clause (iv); (iii) [sign density, conditional] conditional on the Gaussian fluctuation model for the normalized remainder in (i)–(ii), the logarithmic density of $\{D_1 > 0\}$ exists and equals $\frac{1}{2}$, approached from above with a finite- x excess of order $1/\log x$ (the drift-to-noise ratio $T/\sqrt{\pi_t} \asymp 1/\log x$): in contrast to Chebyshev’s races, the twin race has no persistent leader; (iv) [mod-8 companion] since $q^2 - 2 \equiv 7 \pmod{8}$ always (Lemma 1), the entire deterministic term falls on the single class 7 (mod 8): with $D_7(x) = \frac{1}{3} \sum_{a \in \{1, 3, 5\}} \pi_t(x; 8, a) - \pi_t(x; 8, 7)$,

$$\mathcal{M}_x\left(\frac{D_7 - 2T}{\sqrt{\pi_t}}\right) \rightarrow 0$$

(twice the mod-5 term, undiluted), while classes 1, 3, 5 are mutually symmetric. The mod-5 and mod-8 races are two projections of one mechanism, so their joint behaviour is a family test in the sense of the shape-of-agreement criterion.

An earlier draft of clause (iii) asserted a limiting sign density strictly between $\frac{1}{2}$ and 1; an external review correctly observed that this is inconsistent with the scale analysis—when drift/noise $\rightarrow 0$, a Gaussian model forces the sign density to $\frac{1}{2}$, and a nontrivial limiting density would require a persistent normalized bias that no mechanism here produces. A yet earlier draft wrote the error in (iv) as $O^*(\sqrt{\pi_t})$; the fourth review correctly objected that O -notation asserts a provable bound where only a conjectural fluctuation statement is meant, and the $\mathcal{M}_x$ -clauses above say exactly what is conjectured and no more. Two model assumptions remain open and are stated as such: the passage from the Λ -weighted count to unweighted twin counts is a partial-summation argument whose error terms need writing out (higher prime powers are disposed of by Lemma 1); and the *zero-oscillation caveat*—the drift law treats the oscillatory explicit-formula contributions to class differences as having vanishing logarithmic mean, the exact analogue of the Rubinstein–Sarnak GRH-plus-linear-independence framework [6], and a conspiracy among hypothetical zeros could in principle mimic or cancel part of the drift; clause (i) is conjectured on the standard side of that hypothesis.

The mechanism is the twin analogue of the classical explanation of Chebyshev’s bias: prime-square terms contaminate the classes that can contain them. (Biases of twin primes in residue classes have been studied before—Sahoo [25] for the mod-4 race, and [26] for prime pairs versus isolated primes—but we could find no prior statement of the mod-5 race, the $(q^2 - 2, q^2)$ mechanism, or the quantified surplus below.) Mod 3 kills the pattern $(q^2, q^2 + 2)$ outright, an amusing local accident that halves the mechanism and (as a mod-3 computation showed, overturning the engine’s first guess of a quadratic-residue bias against class 4) concentrates the surplus entirely on class 1. Data to $4 \cdot 10^9$: class counts (3980017, 3982505, 3981922), D_1 within one noise unit of T throughout, no persistent leader, and the control race $\pi_{t,2} - \pi_{t,4}$ wandering like a fair coin. Mod-8 data to 10^9 : class counts (856684, 855807, 856046, 855967) for $a = 1, 3, 5, 7$; $D_7 = +212$ at 10^9 against predicted $T = +254$ with noise ~ 1068 , positive through most of the

range (log-density 0.775), and the three pairwise $\{1, 3, 5\}$ controls all within one noise unit of zero—the two projections of the mechanism behave coherently, none of it provable at present.

Conjecture 22 (Least primes in progressions; (a) for part (i) [18, 19, 28, 27], (b) for part (ii)). *Let $p(a, q)$ be the least prime $\equiv a \pmod{q}$ and $U(a, q) = \text{Li}(p(a, q))/\varphi(q)$. Then: (i) $U \Rightarrow \text{Exp}(1)$ marginally as $q \rightarrow \infty$; and, under the additional hypothesis that the class processes are asymptotically jointly independent in the extreme-value sense (which marginal convergence alone does not supply), $\max_a U - H_{\varphi(q)}$ converges to a Gumbel law ($H_n = \sum_{k \leq n} 1/k$), the order-statistics form of Wagstaff’s $\max_a p(a, q) \sim \varphi(q) \log^2 q$; (ii) the finite- q deficit, averaged over all reduced classes a at fixed q , decomposes at first order $1/\log q$ as*

$$1 - \mathbb{E}_a[U] = \frac{\theta_{\text{disc}}(q) + \theta_{\text{corr}}(q)}{\log q} + o\left(\frac{1}{\log q}\right),$$

where $\theta_{\text{disc}}(q)$ is now defined by formula from a named control (the parity-aware Cramér–Bernoulli model: candidates $a, a + q, a + 2q, \dots$ are independently “prime” with hazard $h_i = (q/\varphi(q)) \cdot s_i / \log(a + iq)$, $s_i = 2$ for odd candidates and 0 for even, matching density and parity; $\theta_{\text{disc}}(q) = (1 - \mathbb{E}_a[U^{\text{model}}]) \log q$ is computable exactly for each q), and $\theta_{\text{corr}}(q) = (1 - \mathbb{E}_a[U]) \log q - \theta_{\text{disc}}(q)$ is the ordering term—the part not explained by discreteness. The conjecture is that $\theta_{\text{corr}}(q) > 0$ persists as $q \rightarrow \infty$ along prime moduli, where smooth-modulus singular-series structure is absent.

The stratified experiment that clause (ii) now rests on—demanded in substance by the fourth external review—separates moduli by factorization type in the matched range $q \in [1500, 6000]$: over 40 *prime* moduli, $\theta_{\text{disc}} = 0.847$ (essentially constant across q) and $\theta_{\text{corr}} = 0.824 \pm 0.009$; over 40 *7-smooth* moduli in the same range, $\theta_{\text{disc}} = 4.32 \pm 0.22$ (the large $q/\varphi(q)$ hazards make discreteness dominant) while $\theta_{\text{corr}} = -2.32 \pm 0.22$ reverses sign. The comparison with Leung [27], raised by the third review, is thereby given its answer-shaped experiment: Leung derives the exponential law under a uniform Hardy–Littlewood hypothesis and reports discrepancies for *smooth* moduli, and our smooth stratum indeed behaves anomalously—but the positive ordering term survives, with small error bars, exactly where smooth-modulus effects cannot reach. Whether θ_{corr} tends to a constant, and the exact functional form in $1/\log q$, remain open: data at $q \leq 6000$ cannot distinguish $1/\log q$ from $(\log \log q)/\log q$ -type refinements—the aggregate $\theta \approx 1.5$ – 1.9 over $q \in [10^2, 6 \times 10^3]$, with $\mathbb{E}_a[U]$ bottoming near 0.737 around $q \approx 200$ and recovering through 0.776 by $q \approx 6000$, is calibration, not a determination of the form. The averaging family is now fixed as: all moduli of the stated type in dyadic ranges of q , per stratum.

Remark 1. *The deficit is reproducible and is not an artifact of the time-change: a Cramér pseudo-prime control (independent Bernoulli “primes” with the correct densities and parity) shows roughly half of it—the discreteness of large per-candidate hazards—while the other half vanishes under any reshuffling of the actual prime residues (iid labels, or a random permutation of the true residue sequence, both restore $\mathbb{E}[U] = 1.00$). The ordered sequence of primes fills residue classes measurably faster than any exchangeable model, and the stratified experiment above shows the effect is not a smooth-modulus artifact: it is largest and cleanest on prime moduli. We measure θ_{corr} ; we cannot yet derive it. It is the deepest unexplained number produced by this exercise.*

Conjecture 23 (Fermat quotients and the Wieferich ledger; (a), cf. [20]). *With $q_p = (2^{p-1} - 1)/p \pmod{p}$, three logically independent clauses: (i) [global equidistribution] q_p/p equidistributes on $[0, 1)$ with discrepancy at the calibrated random-model scale: $D_{\text{KS}}(x) = O(\sqrt{\log \log \pi(x)/\pi(x)})$*

(the Chung–Smirnov law-of-the-iterated-logarithm scale; an earlier draft demanded $O(\pi(x)^{-1/2})$, which is stronger than the random model supports—the same over-demand that killed the Mertens conjecture, caught here by external review), with no drift; (ii) [shrinking targets] for every fixed $K \geq 1$, and uniformly for $K = K(x) \leq \log x$, $\#\{p \leq x : q_p < K\} \sim \sum_{p \leq x} \min(1, K/p)$ —the fixed- K case down to $K = 1$ is what governs the zero event and does NOT follow from (i), since $\{q_p = 0\}$ is a target of shrinking measure $1/p$; (iii) [Wieferich count, fluctuation-calibrated] $W(x) = \#\{\text{Wieferich } p \leq x\} = \log \log x + O_\varepsilon((\log \log x)^{1/2+\varepsilon})$, the fluctuation scale being that of the independent-Bernoulli model (variance $\sim \log \log x$).

Two corrections from external review are absorbed here. First, an earlier draft derived (iii) from (i) with “hence”; that inference is invalid—a square-root-size error in the aggregate distribution is astronomically larger than $\log \log x$, and only the shrinking-target clause (ii) at bounded K speaks to the zero event. Second, the earlier fluctuation claim $O(1)$ violated this paper’s own calibration principle (criterion of demanding exactly the fluctuations the model earns): with event probabilities $1/p$ the count has standard deviation $\sim \sqrt{\log \log x}$, and the statement now says so. We also retract any location prediction for a third Wieferich prime: the expected count reaching 3 “at doubly exponential heights” is an expected-count statement whose additive constant is uncalibrated, not a forecast. Consistency data: $\{1093, 3511\}$ below 10^8 against expected ≈ 2.7 ; $\sqrt{n}$ KS = 0.82, 0.82, 0.52, 0.63 at $x = 10^5, \dots, 10^8$ (bounded, no drift); census at $K = 100$: observed 169 against model 161.2.

Conjecture 24 (Conjecture F as a family; (a), [1], cf. [21]). *For odd A let $Q_A(N) = \#\{n \leq N : n^2 + n + A \text{ prime}\}$ and $C(A) = C(x^2 + x + A)$. Then, for every fixed $B > 0$, $Q_A(N) = C(A) I_A(N) (1 + o(1))$ uniformly over odd $1 \leq A \leq (\log N)^B$, with residuals $z_A(N) = (Q_A - C(A) I_A) / \sqrt{C(A) I_A}$ obeying the analogue of Conjecture 16(ii) with the kernel now derived: two members share the index set $n \leq N$, so*

$$\text{Cov}(Q_A, Q_{A'}) \sim [C(A, A') - C(A) C(A')] I_{A, A'}(N), \quad C(A, A') = C(x^2 + x + A, x^2 + x + A'),$$

with $C(A, A') = 0$ (hence negative correlation) for the locally exclusive pairs, and the residual field converges to a centered Gaussian field with the corresponding correlation kernel. (Uniformity over a range growing with N is the substantive claim; as an external review noted, uniformity over any fixed finite set of A is logically automatic from the individual asymptotics.) Among odd $A \leq 199$, Euler’s $A = 41$ has the largest constant, $C(41) = 6.6395463 \dots$ (Cohen’s high-precision value [29]).

Numerical evaluation of the kernel (all odd $A, A' \leq 99$ at $N = 10^6$: 1,225 pairs, of which 289 locally exclusive) returned a structurally clean answer: the positive correlations of admissible pairs and the negative correlations of exclusive pairs *cancel*, mean off-diagonal $\rho = -0.003$, so the derived cross-member kernel is null and the members are asymptotically independent after centering. The observed profile variance ≈ 0.37 over the same range is therefore *not* a common-mode effect; it must be diagonal—each $\text{Var}(Q_A)$ sub-Poisson through within-sequence pair correlations, the quadratic-family analogue of the Montgomery–Soundararajan deficit (Conjecture 20)—and part of the spread is truncation noise in the constants themselves (the $\pm 10^{-3}$ wobble moves each z_A by ~ 0.3). Deriving the diagonal law is the family’s remaining open component; the cross-member question the review posed is now closed by computation. At $N = 10^6$: mean ratio 1.0006, standard deviation 0.0027, max $|z| = 1.92$, rank correlation 0.99988; for $A = 41$, observed 261,080 against predicted 261,017.6 ($z = +0.12$, using Cohen’s constant).

A cautionary note from the referee audit belongs here: our own truncated Euler product for $C(41)$ at cutoff $2 \cdot 10^5$ reads 6.64092—wrong in the fourth significant decimal—because these conditionally convergent products drift at the 10^{-3} scale over practical cutoffs. Quoted digits for all quadratic-family constants in this paper are limited by the stated truncation wobble, not by the last printed digit. The entire 10^8 -bit primality dataset compresses to one product formula per A —the description-length virtue in its purest form.

Conjecture 25 (The Goldbach lane race; apparently new). *For $n \equiv 2 \pmod{4}$ let $R_1(n)$, $R_3(n)$ be the ordered Goldbach representation counts in the lanes $p \equiv q \equiv 1$ and $p \equiv q \equiv 3 \pmod{4}$ (Conjecture 15), and $D(n) = R_3(n) - R_1(n)$. Prime-square terms $n = q^2 + p$ (q odd prime) land exclusively in the (1,1) lane, since $q^2 \equiv 1 \pmod{4}$ and then $n - q^2 \equiv 1 \pmod{4}$ automatically. Consequently the (3,3) lane leads on average: with*

$$D_{\text{sys}}(n) = \frac{2}{\bar{\ell}(n)} \sum_{\substack{q \text{ odd prime, } q \leq \sqrt{n} \\ n - q^2 \text{ prime}}} \log q \log(n - q^2), \quad \bar{\ell}(n) = \frac{n - 6}{\int_3^{n-3} \frac{dt}{\log t \log(n - t)}},$$

(the analytic mean of $\log p \log(n - p)$ under the lane profile; the empirical lane mean is its estimator). Write $\mathbb{E}_x$ for the average over n drawn from $\{n \equiv 2 \pmod{4}, n \leq x\}$ with weight proportional to $1/n$ (the logarithmic sampling measure, now explicit). Then: (i) [drift law] $\mathbb{E}_x[D - D_{\text{sys}}] = o(\mathbb{E}_x[D_{\text{sys}}])$ as $x \rightarrow \infty$: the (3,3) lane leads on average by exactly the square contamination; (ii) [weighted mean and internal null lane] $\mathbb{E}_x[D_{\text{sys}}]$ is given by the Hardy–Littlewood (Conjecture H) local model for $n - q^2$, whose local factors vary with n as the fourth external review stressed—most drastically at $p = 3$: for $n \equiv 1 \pmod{3}$, $3 \mid n - q^2$ for every prime $q > 3$, so the contamination collapses to the single $q = 3$ term and the race carries essentially no drift on the subprogression $n \equiv 10 \pmod{12}$, while the drift concentrates on $n \not\equiv 1 \pmod{3}$ —an internal null lane, predicted by the same mechanism that predicts the lead; (iii) [sign density] the per- n drift-to-noise ratio $\kappa(n) = D_{\text{sys}}(n) / \sqrt{\mathfrak{S}(n)J(n)}$ (where $J(n) = \int_3^{n-3} dt / (\log t \log(n - t))$, so that $\mathfrak{S}(n)J(n)$ is the total ordered representation count) satisfies $\kappa(n) \asymp c(n) / \log n \rightarrow 0$; under the Gaussian fluctuation model the logarithmic density of $\{D > 0\}$ is therefore $\frac{1}{2}$ in the limit, with the finite- x sign fraction predicted to be $\mathbb{E}_x[\Phi(\kappa)]$ (Φ the standard normal distribution function)—decaying, but far from $\frac{1}{2}$ at any accessible height.

Derivation status: the identity behind D_{sys} is the Λ -weighted lane symmetry minus its prime-power part; the transfer to unweighted counts is a partial-summation argument whose error terms (higher prime powers, which enter only at the $n^{1/3}$ scale, and the variation of the weight across the lane) remain to be written out—the same open transfer flagged at Conjecture 21—and clause (iii) inherits the zero-oscillation caveat stated there.

This is the Goldbach-partition analogue of Chebyshev’s bias, produced by the same explicit-formula mechanism as Conjecture 21 but through $q^2 + p$ patterns rather than twin patterns; the single-sign prediction (every square contamination falls into one lane) makes it cleaner than the classical race. We could find no prior statement. It replaces, at the suggestion of an external review, a restatement of the Lemke Oliver–Soundararajan mod-3 law that formerly occupied this slot and carried no new content (their conjecture remains cited at Conjecture 16’s family level and its verification data— $\hat{c}$ drifting $0.400 \rightarrow 0.370$ over $10^7 \rightarrow 4 \cdot 10^9$ with (1,1)/(2,2) symmetry 0.99989—remain in the repository).

Verification. Over 500 log-spaced samples $n \leq 10^8$: mean $D = 76.5$ against mean empirical $D_{\text{sys}} = 64.5$, the (3,3)-lane lead significant at 5.2 standard errors of the mean. Clause (ii)’s

weighted model, computed in presieve-exact form (trial division of each $n - q^2$ to 10^3 with the exact Mertens normalization), reproduces the empirical contamination with no fitted parameter: model mean 66.3 against empirical 67.8 (ratio 0.98) on a 400-sample run. The internal null lane behaves as predicted: on $n \equiv 1 \pmod{3}$ (123 samples) the model and empirical D_{sys} both vanish and the measured lead drops to 31 ± 23 (consistent with zero), while on $n \not\equiv 1 \pmod{3}$ (277 samples) the lead is 88 ± 21 against predicted contamination 96—the drift lives exactly where the mechanism puts it. Clause (iii): computed $\mathbb{E}_x[\Phi(\kappa)] = 0.572$ (κ ranging 0–0.64 with mean 0.18) against observed sign fraction 0.588; weak, and quantitatively the weakness the model requires.

7 The adversarial audit

Each statement above was attacked three ways.

Layer 1: literature. Five independent search agents combed OEIS, arXiv, and the standard references for prior art, producing the novelty labels and the attribution section below. This layer also falsified one historical claim in the engine’s working notes (the $k \geq 1$ Stern list of Conjecture 14 was described as partly new; it is OEIS A060003 verbatim, and 1493 is the largest known Stern prime); the notes were corrected on the record. The layer also produced the audit programme’s own worst failure: a snippet-depth search cleared the primorial-twin statement that Lillie [5] had in fact already made (Finding 2). All novelty searches for the statements admitted in the present revision (the Conjecture 1 and 5 family lifts, the chain of Conjecture 6, the null race of Conjecture 8, the factorial twins of Conjecture 10) were therefore run at abstract depth: the nearest-neighbour papers were read, not skimmed from search results.

Layer 2: scripted refutation. Every falsifiable-by-instance statement was pushed at least a decade past its original bound: exception hunts across $(10^8, 10^9]$ for Conjectures 13, 14, 15, 17; the uniformity of Conjecture 16 stressed on $d \leq 6000$; the statistical laws re-tested at $4 \cdot 10^9$ (Conjectures 18–21, 25, and the quintuplet count of Conjecture 4); the recovery clause of Conjecture 22 tested on fresh moduli $q \in (3000, 6000]$. Outcome: one kill (Finding 3); no other statement moved. The statements admitted or upgraded in the present revision were verified at production scale before admission: the Conjecture 1 profile across 150 shifts; the chain of Conjecture 6 over five decades to 10^7 ; the null race of Conjecture 8 at 10^7 ; factorial twins to $n = 700$; the CRT-exact factorization of Conjecture 11 through a joint period of 1.2×10^8 with the scan extended to $n = 6000$; the covariance kernels of Conjectures 16(ii) and 24(ii) evaluated over 870 and 1,225 pairs; the stratified experiment of Conjecture 22 on 80 moduli; and the weighted drift and internal null lane of Conjecture 25 on 400 fresh samples.

Layer 3: clean-context replication. Independent agents, given only the bare statements and forbidden to read the engine’s code or notes, recomputed constants from definitions and recounted sequences with their own implementations. All recomputed constants landed inside the stated truncation error; all recounts agreed (one replication initially disagreed at $z \approx -22$, traced by the replicating agent itself to a bug in *its* sieve—every prime $n \leq Q$ had been struck out by its own modulus—after which it agreed at $z < 1.6$). The same layer reproduced the Fibonacci deficit of Remark after Conjecture 9 and the ordering anomaly of Remark 1, confirming both are properties of the primes rather than of our code.

We draw one methodological conclusion. Every fatal mathematical error the audit programme has caught in its own output was not statistical but *algebraic*: factorizations or congruence collapses living on density-zero or positive-density-but-structured subsequences (Findings 3 and 1, the $k = 4$ parity branch of Conjecture 5, the inadmissible naive chain of Conjecture 6); and the

one fatal *bibliographic* error was a search read at snippet depth (Finding 2). Probabilistic sanity checks, however extensive, integrate over density and cannot see such families; only structural attack surfaces (here: “test the claimed exception census on special subsequences”) find them. An audit that only re-runs the original verification at a larger bound would have declared the false version of Conjecture 13 sound.

8 Attribution and novelty

Previously stated in essence, with our contribution being the uniform quantification, computed constants, and re-verification: C1_{*d*=2} (OEIS A080149; [30]); C3 [10]; C6’s length-2 sub-chain (constant catalogued as OEIS A188596); C9 [7]; C12 [4]; C13-core [1]; C14 (OEIS A060003); C15 [11]; C16-core (uniform Hardy–Littlewood B, cf. [12]); C17 [9]; C18 [13]; C23 [20]; C24-core [1, 21]. Retired former occupants, previously stated: the primorial twins of slot 10, Lillie [5] (see Finding 2); the primorial benchmark before that and the Lemke Oliver–Soundararajan restatement of slot 25, [4] and [22] (the third external review caught a stale attribution table for these, corrected on the record); the Sophie Germain statement of slot 6 and the quadratic triple of slot 8, retained inside the new statements as attributed calibration.

For the remainder we distinguish, following the external review, two grades of novelty. *Bibliographic novelty only*—unrecorded specializations of Bateman–Horn, analogous to evaluating a classical special function at a new argument, offered as benchmark data: the constants of C2, C5(iii) and C7. *Conceptual novelty*—a mechanism, law, family uniformity, or dichotomy we could not find in the literature: C1’s uniform quadratic de Polignac family (the profile law, not the $d = 2$ instance); C4’s power-obstruction ladder; C5’s complete twin-base family analysis; C6’s alternating cyclotomic chain; C8’s null-mechanism contrast prediction; C10’s factorial-twin finiteness (the convergent Borel–Cantelli exemplar, admitted after the deeper search of Finding 2); C11’s entanglement-aware local product for a mixed polynomial–exponential sequence; C16(ii)’s derived covariance kernel (and its C24(ii) family analogue with the computed null cross-kernel); C19’s liminf constant (against the slope-1 laws of Shanks, Wolf, and Kourbatov–Wolf [23]); C20’s microscopic variance law (a sharpening of [17], with the mesoscopic numerics of [24] as precedent); C21’s race mechanism (against the twin-bias literature [25, 26]); C22(ii)’s deficit decomposition with its stratified differentiating experiment; and C25’s Goldbach lane race. Among these we consider C21, C25 and C8 (the square-contamination mechanism family with its positive and negative controls), C22(ii) (the least-prime deficit law with its ordering anomaly), and the derived kernels of C16(ii) and C24(ii) the paper’s most genuinely new content, and Question 1 its best export.

Question 1. Determine $\theta(q)$ of Conjecture 22: derive, from the Hardy–Littlewood correlations or otherwise, the first-order deficit of $\mathbb{E}_a \text{Li}(p(a, q))/\varphi(q)$ below its random-model value 1—equivalently, explain quantitatively why the primes in their natural order occupy residue classes faster than any exchangeable resampling of themselves.

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